

Comparing and Ordering fractions

When we compare fractions, we can use a fraction bar to help us to see which fraction is bigger. For example:

Which is bigger: $\frac{3}{4}$ or $\frac{4}{5}$?

Draw two bars with the same area.



$\frac{3}{4}$



$\frac{4}{5}$

By comparing the shaded parts, we can see that $\frac{4}{5}$ is bigger than $\frac{3}{4}$.

We can also compare fractions by finding a **common denominator**. In order to find a common denominator, we need to find the lowest common multiple of the denominators of the fractions we are comparing. Let's check the multiples of 4 and 5 to find the lowest multiple they share:

4, 8, 12, 16, **20**, 24, 28, 32...

5, 10, 15, **20**, 25, 30, 35, 40...

We can see that we can use the number **20** as the denominator. Next, we need to find out what we have multiplied the denominator in each fraction by to get to 20, then do the same to the numerator:

$$\begin{array}{ccc} \frac{3}{4} & & \frac{4}{5} \\ \times 5 \swarrow & & \searrow \times 4 \\ \frac{15}{20} & & \frac{16}{20} \end{array}$$

We can see that $\frac{16}{20}$ is bigger than $\frac{15}{20}$, so let's go back to the original question:

Which is bigger: $\frac{3}{4}$ or $\frac{4}{5}$?

$\frac{16}{20}$ is equivalent to $\frac{4}{5}$, so we know that $\frac{4}{5}$ is bigger than $\frac{3}{4}$.

Compare the fractions

Find the bigger fraction in each pair. Find the lowest common multiple to help you to compare the fractions. The first one has been done for you.

Which is bigger:

$\frac{3}{5}$ or $\frac{7}{8}$?

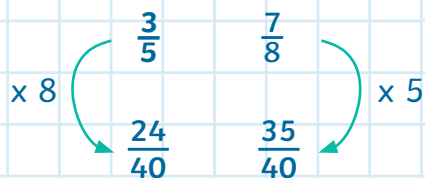
Multiples of 5:

5, 10, 15, 20, 25, 30, 35, **40**...

Multiples of 8:

8, 16, 24, 32, **40**, 48, 56, 64...

Lowest common multiple = **40**



$\frac{7}{8}$ is bigger.

Which is bigger:

$\frac{5}{8}$ or $\frac{4}{6}$?

is bigger.

Which is bigger:

$\frac{2}{3}$ or $\frac{4}{11}$?

is bigger.

Which is bigger:

$\frac{9}{10}$ or $\frac{7}{8}$?

is bigger.

Which is bigger:

$\frac{5}{6}$ or $\frac{7}{9}$?

is bigger.

Which is bigger:

$\frac{5}{12}$ or $\frac{4}{7}$?

is bigger.

Ordering Fractions

Order the fractions from smallest to largest.

$\frac{7}{8}$	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{5}{6}$
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.....

Multiples of 8: 8, 16, **24**, 32, 40, 48, 56, 64...

Multiples of 3: 3, 6, 9, 12, 15, 18, 21, **24**, 27...

Multiples of 2: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, **24**...

Multiples of 6: 6, 12, 18, **24**, 30, 36, 42, 48...

.....

The lowest common multiple is **24**.

$$\begin{array}{ccc} & \frac{7}{8} & \\ \text{x } 3 \swarrow & & \searrow \text{x } 3 \\ & \frac{21}{24} & \end{array}$$

$$\begin{array}{ccc} & \frac{2}{3} & \\ \text{x } 8 \swarrow & & \searrow \text{x } 8 \\ & \frac{16}{24} & \end{array}$$

$$\begin{array}{ccc} & \frac{1}{2} & \\ \text{x } 12 \swarrow & & \searrow \text{x } 12 \\ & \frac{12}{24} & \end{array}$$

$$\begin{array}{ccc} & \frac{5}{6} & \\ \text{x } 4 \swarrow & & \searrow \text{x } 4 \\ & \frac{20}{24} & \end{array}$$

.....

Now, we can order them:

Smallest			Largest
$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	$\frac{7}{8}$

Order the Fractions

Order each set of fractions from smallest to largest. Remember to find the lowest common multiple to help.

$\frac{5}{6}$

$\frac{7}{9}$

$\frac{2}{3}$

Smallest

Largest

$\frac{3}{4}$

$\frac{3}{5}$

$\frac{7}{10}$

Smallest

Largest

$\frac{6}{15}$

$\frac{3}{10}$

$\frac{7}{20}$

Smallest

Largest

$\frac{5}{6}$

$\frac{2}{5}$

$\frac{7}{15}$

$\frac{1}{2}$

$\frac{23}{30}$

Smallest

Largest

$\frac{19}{24}$	$\frac{7}{12}$	$\frac{5}{16}$	$\frac{5}{6}$	$\frac{3}{8}$
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Smallest

Largest

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$\frac{5}{8}$	$\frac{5}{6}$	$\frac{3}{4}$	$\frac{7}{9}$	$\frac{2}{3}$
---------------	---------------	---------------	---------------	---------------

Smallest

Largest

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Challenges

1. Dawson says, "I know that $\frac{3}{8}$ must be smaller than $\frac{7}{14}$ without changing the denominators."

How does he know?

2. Radhika compares some fractions to order them.

$\frac{7}{9}$	$\frac{5}{6}$	$\frac{3}{4}$
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This is her working.

$$\begin{array}{c} \frac{7}{9} \\ \times 4 \quad \times 4 \\ \hline \frac{28}{36} \end{array}$$

$$\begin{array}{c} \frac{5}{6} \\ \times 6 \quad \times 6 \\ \hline \frac{20}{36} \end{array}$$

$$\begin{array}{c} \frac{3}{4} \\ \times 9 \quad \times 9 \\ \hline \frac{27}{36} \end{array}$$

Smallest

Largest

$\frac{5}{6}$	$\frac{3}{4}$	$\frac{7}{9}$
---------------	---------------	---------------

What has she done wrong?

3. Arlo has 120 sweets. He gives $\frac{7}{30}$ of the sweets to Joey, $\frac{1}{12}$ to Hana and $\frac{11}{40}$ to Ty. How many sweets does he have left?

Comparing and Ordering Fractions

Answers

Compare the Fractions

Which is bigger:

$\frac{3}{5}$ or $\frac{7}{8}$?

.....

Multiples of 5:

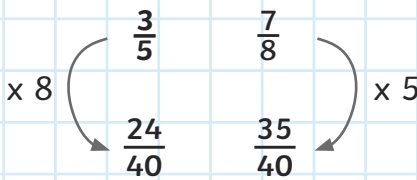
5, 10, 15, 20, 25, 30, 35, **40**...

Multiples of 8:

8, 16, 24, 32, **40**, 48, 56, 64...

.....

Lowest common multiple = **40**



$\frac{7}{8}$ is bigger.

Which is bigger:

$\frac{5}{8}$ or $\frac{4}{6}$?

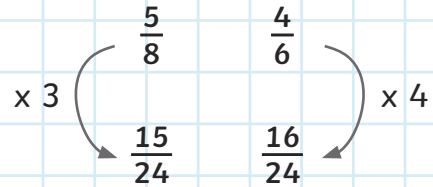
Multiples of 8:

8, 16, **24**, 32, 40, 48, 56, 64...

Multiples of 6:

6, 12, 18, **24**, 30, 36, 42, 48...

Lowest common multiple = **24**



4

6

is bigger.

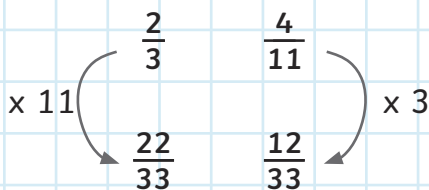
Which is bigger:

$\frac{2}{3}$ or $\frac{4}{11}$?

Multiples of 3: 3, 6, 9, 12, 15, 18, 24, 27, 30, **33**, 36...

Multiples of 11: 11, 22, **33**, 44, 55...

Lowest common multiple = **33**



2

3

is bigger.

Which is bigger:

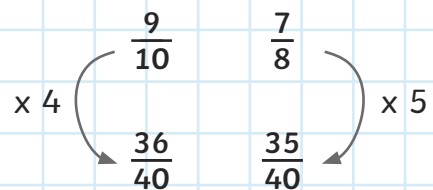
$\frac{9}{10}$ or $\frac{7}{8}$?

Multiples of 10:

10, 20, 30, **40**, 50, 60...

Multiples of 8: 8, 16, 24, 32, **40**, 48...

Lowest common multiple = **40**



9

10

is bigger.

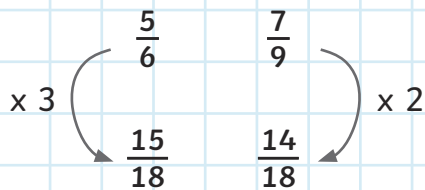
Which is bigger:

$\frac{5}{6}$ or $\frac{7}{9}$?

Multiples of 6: 6, 12, **18**, 24, 30...

Multiples of 9: 9, **18**, 27, 36, 45...

Lowest common multiple = **18**



5

6

is bigger.

Which is bigger:

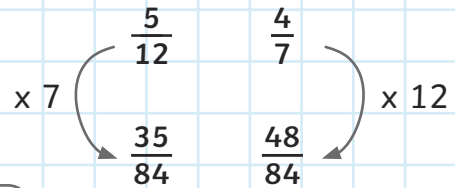
$\frac{5}{12}$ or $\frac{4}{7}$?

Multiples of 12:

12, 24, 36, 48, 60, 72, **84**, 96...

Multiples of 7: 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, **84**, 91...

Lowest common multiple = **84**



4

7

is bigger.

Order the Fractions

$\frac{5}{6}$	$\frac{7}{9}$	$\frac{2}{3}$
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$\frac{5}{6} \xrightarrow{\times 3} \frac{15}{18}$

$\frac{7}{9} \xrightarrow{\times 2} \frac{14}{18}$

$\frac{2}{3} \xrightarrow{\times 6} \frac{12}{18}$

Smallest
Largest

$\frac{2}{3}$	$\frac{7}{9}$	$\frac{5}{6}$
---------------	---------------	---------------

$\frac{3}{4}$	$\frac{3}{5}$	$\frac{7}{10}$
---------------	---------------	----------------

$\frac{3}{4} \xrightarrow{\times 5} \frac{15}{20}$

$\frac{3}{5} \xrightarrow{\times 4} \frac{12}{20}$

$\frac{7}{10} \xrightarrow{\times 2} \frac{14}{20}$

Smallest
Largest

$\frac{3}{5}$	$\frac{7}{10}$	$\frac{3}{4}$
---------------	----------------	---------------

$\frac{6}{15}$	$\frac{3}{10}$	$\frac{7}{20}$
----------------	----------------	----------------

$\frac{6}{15} \xrightarrow{\times 4} \frac{24}{60}$

$\frac{3}{10} \xrightarrow{\times 6} \frac{18}{60}$

$\frac{7}{20} \xrightarrow{\times 3} \frac{21}{60}$

Smallest
Largest

$\frac{3}{10}$	$\frac{7}{20}$	$\frac{6}{15}$
----------------	----------------	----------------

$\frac{5}{6}$	$\frac{2}{5}$	$\frac{7}{15}$	$\frac{1}{2}$	$\frac{23}{30}$
---------------	---------------	----------------	---------------	-----------------

$\frac{5}{6} \xrightarrow{\times 6} \frac{25}{30}$

$\frac{2}{5} \xrightarrow{\times 6} \frac{12}{30}$

$\frac{7}{15} \xrightarrow{\times 2} \frac{14}{30}$

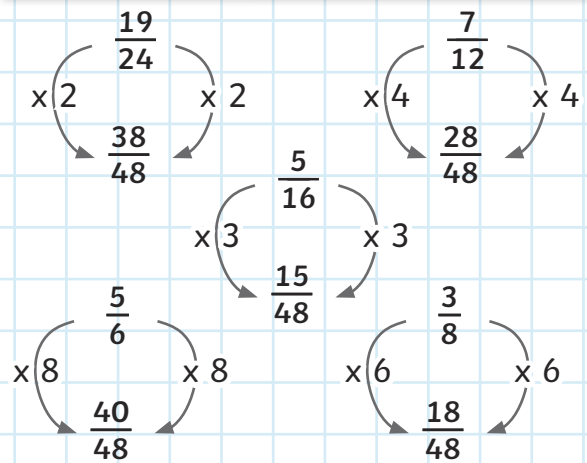
$\frac{1}{2} \xrightarrow{\times 15} \frac{15}{30}$

$\frac{23}{30}$

Smallest
Largest

$\frac{2}{5}$	$\frac{7}{15}$	$\frac{1}{2}$	$\frac{23}{30}$	$\frac{5}{6}$
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$\frac{19}{24}$	$\frac{7}{12}$	$\frac{5}{16}$	$\frac{5}{6}$	$\frac{3}{8}$
-----------------	----------------	----------------	---------------	---------------

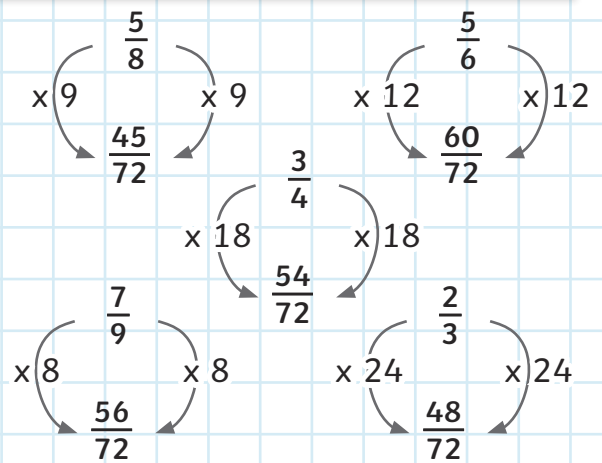


Smallest

Largest

$\frac{5}{16}$	$\frac{3}{8}$	$\frac{7}{12}$	$\frac{19}{24}$	$\frac{5}{6}$
----------------	---------------	----------------	-----------------	---------------

$\frac{5}{8}$	$\frac{5}{6}$	$\frac{3}{4}$	$\frac{7}{9}$	$\frac{2}{3}$
---------------	---------------	---------------	---------------	---------------



Smallest

Largest

$\frac{5}{8}$	$\frac{2}{3}$	$\frac{3}{4}$	$\frac{7}{9}$	$\frac{5}{6}$
---------------	---------------	---------------	---------------	---------------

Challenges

1. Dawson says, "I know that $\frac{3}{8}$ must be smaller than $\frac{7}{14}$ without changing the denominators."

How does he know?

$\frac{7}{14}$ is the same as $\frac{1}{2}$ and $\frac{3}{8}$ is smaller than a half, so it must be smaller than $\frac{7}{14}$.

2. Radhika compares some fractions to order them.

$\frac{7}{9}$	$\frac{5}{6}$	$\frac{3}{4}$
---------------	---------------	---------------

This is her working.

$$\begin{array}{c} \frac{7}{9} \\ \times 4 \quad \times 4 \\ \hline \frac{28}{36} \end{array}$$

$$\begin{array}{c} \frac{5}{6} \\ \times 6 \quad \times 6 \\ \hline \frac{20}{36} \end{array}$$

$$\begin{array}{c} \frac{3}{4} \\ \times 9 \quad \times 9 \\ \hline \frac{27}{36} \end{array}$$

Smallest

Largest

$\frac{5}{6}$	$\frac{3}{4}$	$\frac{7}{9}$
---------------	---------------	---------------

What has she done wrong?

Radhika has made a mistake with the fraction $\frac{5}{6}$. Instead of multiplying the numerator 5 by 6 to get 30, she has multiplied it by 4 to get 20. Instead of being $\frac{20}{36}$, the fraction should be $\frac{30}{36}$. Using this fraction, the actual order from smallest to largest would be:

Smallest

Largest

$\frac{3}{4}$	$\frac{7}{9}$	$\frac{5}{6}$
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3. Arlo has 120 sweets. He gives $\frac{7}{30}$ of the sweets to Joey, $\frac{1}{12}$ to Hana and $\frac{11}{40}$ to Ty.
How many sweets does he have left?

$$\begin{array}{c} \frac{7}{30} \\ \swarrow \quad \searrow \\ \times 4 \quad \times 4 \\ \swarrow \quad \searrow \\ \frac{28}{120} \end{array}$$

$$\begin{array}{c} \frac{1}{12} \\ \swarrow \quad \searrow \\ \times 10 \quad \times 10 \\ \swarrow \quad \searrow \\ \frac{10}{120} \end{array}$$

$$\begin{array}{c} \frac{11}{40} \\ \swarrow \quad \searrow \\ \times 3 \quad \times 3 \\ \swarrow \quad \searrow \\ \frac{33}{120} \end{array}$$

$$28 + 10 + 11 = 49$$

Arlo has given away 49 sweets.

$$120 - 49 = 71$$

Arlo has 71 sweets left.